

1. Tentukan manakah pasangan angka dibawah yang merupakan solusi dari SPL berikut

$$2x_1 - 4x_2 - x_3 = 1$$

$$x_1 - 3x_2 + x_3 = 1$$

$$3x_1 - 5x_2 - 3x_3 = 1$$

(a) $(3, 1, 1)$

(b) $(3, -1, 1)$

(c) $(13, 5, 2)$

(d) $(\frac{13}{2}, \frac{5}{2}, 2)$

(e) $(17, 7, 5)$

2. Tentukan apakah SPL dibawah memiliki solusi Tunggal, tak hingga atau tdk ada solusi, buatlah graphnya

(a) $3x - 2y = 4$

$$6x - 4y = 9$$

(b) $2x - 4y = 1$

$$4x - 8y = 2$$

(c) $x - 2y = 0$

$$x - 4y = 8$$

3. Determine what a and b are in the SPL below has no solutions, a single solution, and infinite solutions.?

$$2x - 3y = a$$

$$4x - 6y = b$$

4. Suppose that a certain diet calls for 7 units of fat, 9 units of protein, and 16 units of carbohydrates for the main meal, and suppose that an individual has three possible foods to choose from to meet these requirements:

Food 1: Each ounce contains 2 units of fat, 2 units of protein, and 4 units of carbohydrates.

Food 2: Each ounce contains 3 units of fat, 1 unit of protein, and 2 units of carbohydrates.

Food 3: Each ounce contains 1 unit of fat, 3 units of protein, and 5 units of carbohydrates.

Let x , y , and z denote the number of ounces of the first, second, and third foods that the dieter will consume at the main meal. Construct the SPL whose solution tells how many ounces of each food must be consumed to meet the diet requirements

5. For the following equation $y = ax^2 + bx + c$ through the point $(1, 1)$, $(2, 4)$, and $(-1, 1)$. Find the equation

6. Suppose that you want to find values for a , b , and c such that the parabola $y = ax^2 + bx + c$ passes through the points $(1, 1)$, $(2, 4)$, and $(-1, 1)$. Find (but do not solve) a system of linear equations whose solutions provide values for a , b , and c .

7. Suppose you are asked to find three real numbers such that the sum of the numbers is 12, the sum of two times the first plus the second plus two times the third is 5, and the third number is one more than the first. Find (but do not solve) a linear system whose equations describe the three conditions

8. Set up a system of linear equations to represent the network shown in Figure below and solve the system

